CSE331
Automata and Computability

FINAL EXAM
TOTAL MARKS: 50
DURATION: 105 MINUTES

B

There are a total of six problems. You have to solve all of them.

Problem 1 (CO5): Pumping Lemma (5 points)

Let $\Sigma = \{0, 1\}$. Consider the following language.

$$L = \{w \in \Sigma^* : w = 1^a 0^b 0^c 1^d, \text{ where } \mathtt{a} + \mathtt{d} = \mathtt{b} + \mathtt{c} \text{ and } \mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d} \geq 0\}$$

Use the pumping lemma to **demonstrate** that L is not regular language.

Problem 2 (CO3): Designing Context-Free Grammars (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages. Recall that for a string w, $|w|$ denotes the length of w.

$$L_1 = \{w \in \Sigma^* : w \text{ is a odd length palindrome}\}$$
$$L_2 = \{w \in \Sigma^* : \text{length of } w \text{ is multiple of two}\}$$
$$L_3 = \{x000y : x, y \in L_2, |x| = |y|\}$$
$$L_4 = L_1 \cap L_3$$

Now solve the following problems.

(a) **Give** a context-free grammar for the language L_1. (3 points)

(b) **Give** a context-free grammar for the language L_3. (4 points)

(c) **Give** a context-free grammar for the language L_4.(3 point)

Problem 3 (CO4): The CYK Algorithm (5 points)

Apply the CYK algorithm to fill up the table for the string aabaa using the following grammar. Here a and b are terminals, and the rest are variables.

$S \to BA \mid BC$
$A \to AB \mid AC \mid \mathtt{a}$
$B \to CB \mid CC \mid \mathtt{b}$
$C \to CA \mid \mathtt{a}$

1,5 $\{?\}$				
1,4 $\{?\}$	2,5 $\{S, A\}$			
1,3 $\{?\}$	2,4 $\{S, A\}$	3,5 $\{?\}$		
1,2 $\{A, B, C\}$	2,3 $\{?\}$	3,4 $\{S\}$	4,5 $\{A, B, C\}$	
1,1 $\{A, C\}$	2,2 $\{A, C\}$	3,3 $\{B\}$	4,4 $\{A, C\}$	5,5 $\{A, C\}$
a	a	b	a	a

Problem 4 (CO3): Constructing Pushdown Automata (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages.

$$L_1 = \{w \mid w \text{ starts and ends with the different character}\}$$

$$L_2 = \{w \mid \text{the number of } 0\text{s in } w \text{ is not same as the number of } 1\text{s}\}$$

(a) **Give** the state diagram of a pushdown automaton that recognizes L_1. (4 points)

(b) **Give** the state diagram of a pushdown automaton that recognizes L_2. (6 points)

Problem 5 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$\begin{aligned} A &\to 1A \mid 0B \mid 00A \\ B &\to 1B \mid 1C \mid 0A \mid 00B \\ C &\to 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid 0 \mid 1 \end{aligned}$$

(a) **Give** a leftmost derivation for the string 10110010. (3 points)

(b) **Sketch** the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) **Demonstrate** that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)

(d) **Find** a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)

(e) **Desgin** an unambiguous Context Free Grammar for the language represented by the given ambiguous grammar. (1 point)

Problem 6 (CO4): Chomsky Normal Form (10 points)

Answer the following questions.

(a) **List** the productions that violate the conditions of the Chomsky Normal Form (CNF) in the following grammar. (5 points)

$$\begin{aligned} P &\to \mathtt{a} \mid PR \mid \varepsilon \\ Q &\to \mathtt{b}P \mid R \\ R &\to QQ \mid \mathtt{a}T \\ T &\to QT \mid \varepsilon \end{aligned}$$

(b) **Write** down the additional rules that need to be added to the following grammar if the production $A \to \varepsilon$ is removed. (3 points)

$$\begin{aligned} S &\to AA \mid \mathtt{b}B \\ A &\to CB \mid \mathtt{a}A \mid \varepsilon \\ B &\to ABA \mid \mathtt{b} \mid \varepsilon \end{aligned}$$

(c) **Write** down the additional rules that need to be added to the following grammar if all the unit productions are removed. (2 points)

$$\begin{aligned} A &\to 0A1B1 \mid C \\ B &\to 1B \mid B \\ C &\to BC \mid A \mid B \end{aligned}$$